

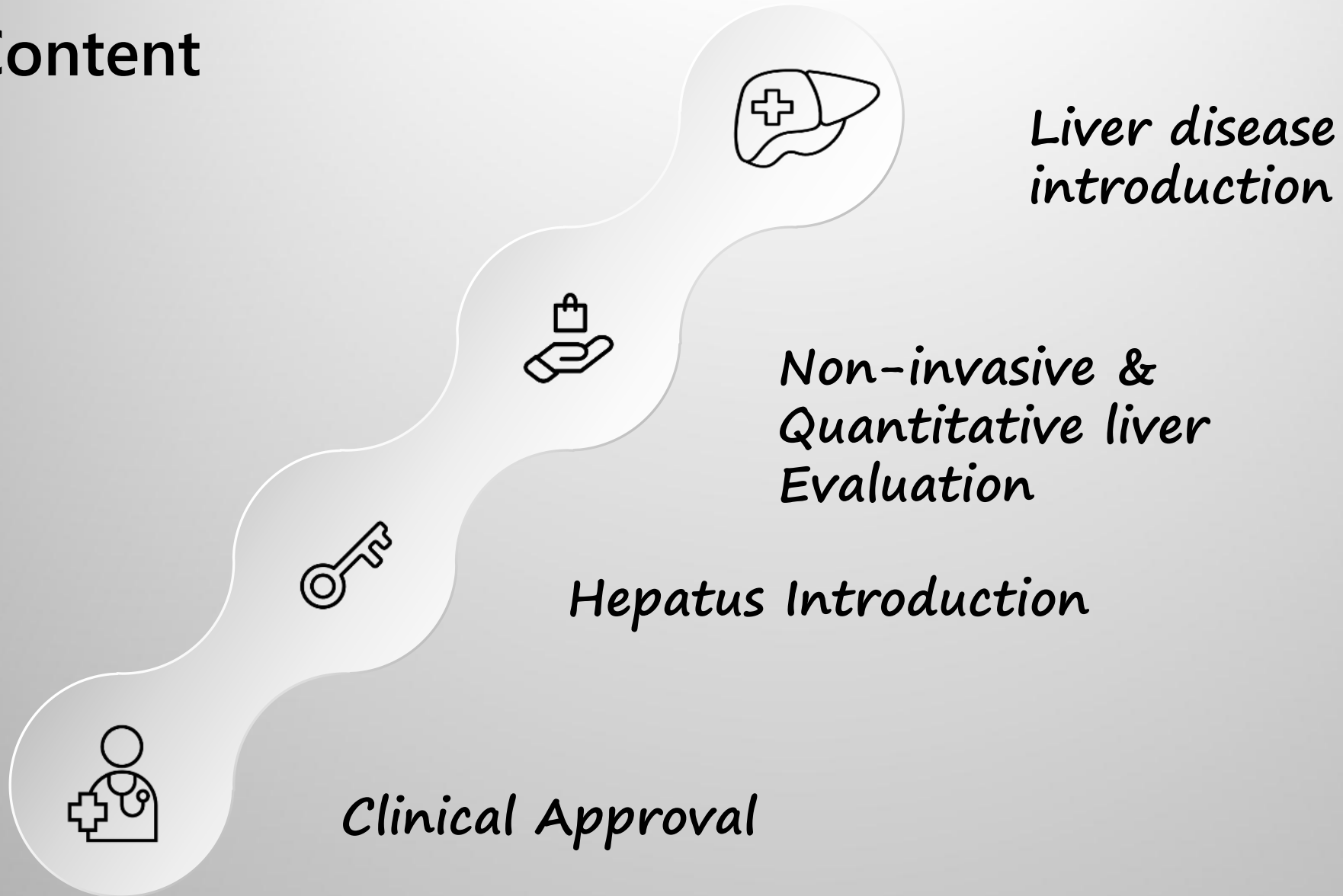


Hepatus 5

*Non-invasive and quantitative
liver diagnose*

*New Vision for Liver
Care*

Content



A photograph of two medical professionals, a man and a woman, in a laboratory or clinical setting. The man, in the foreground, has a grey beard and glasses, wearing a white lab coat over a checkered shirt. The woman, in the background, is smiling and holding a pen, also wearing a white lab coat over a blue striped shirt. They are looking at a laptop screen. The text "Liver disease introduction" is overlaid in red, italicized font.

Liver disease introduction



What is liver disease

- Liver is the metabolic center and the largest detoxification organ of human body
- According to different pathogeny, we can divide liver disease into below types:

Viral hepatitis

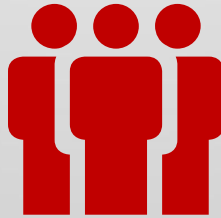
Alcoholic liver disease

Nonalcoholic fatty liver

Metabolic syndrome

Biliary diseases

Autoimmune liver disease



Drug-induced liver injury

Chronic alcoholism

Over nutrition, obesity

hypertension

Diabetes, etc

Liver transplant



Huge population suffer from liver disease



HBV



HCV



NAFLD

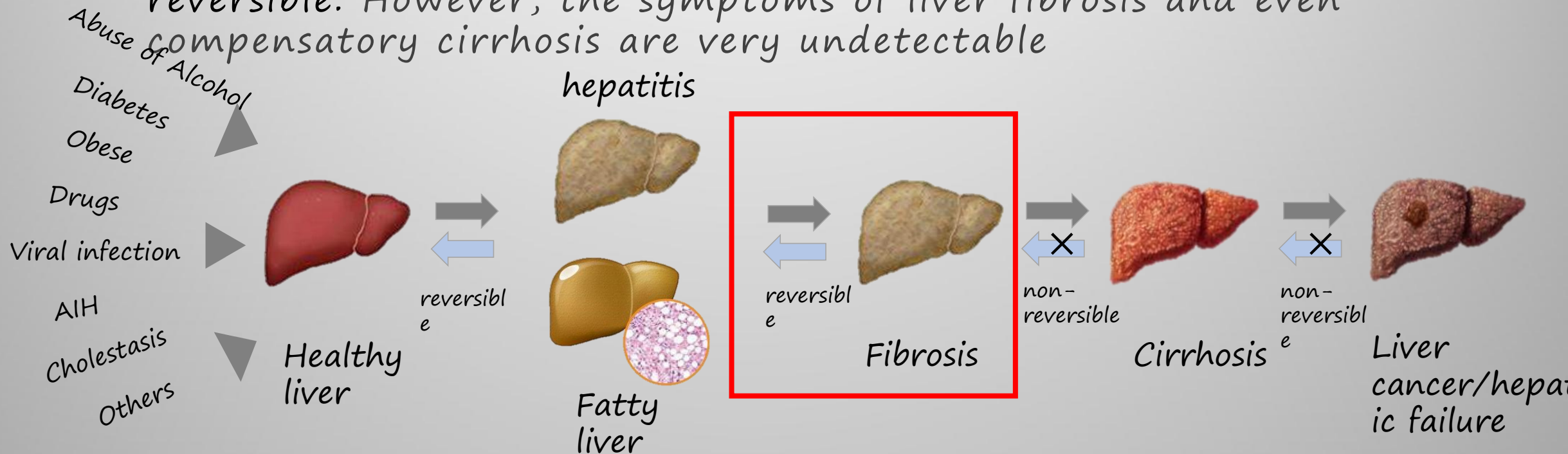


Death
caused by
liver
disease



The evolution of liver disease

- Generally speaking, liver disease has below types and stages: hepatitis/fatty liver → Fibrosis → Cirrhosis → Liver cancer/hepatic failure
- **Fibrosis is reversible**, cirrhosis/liver cancer/ hepatic failure are not reversible. However, the symptoms of liver fibrosis and even compensatory cirrhosis are very undetectable





To prevent fibrosis

Early screening, diagnosis and treatment , **important**



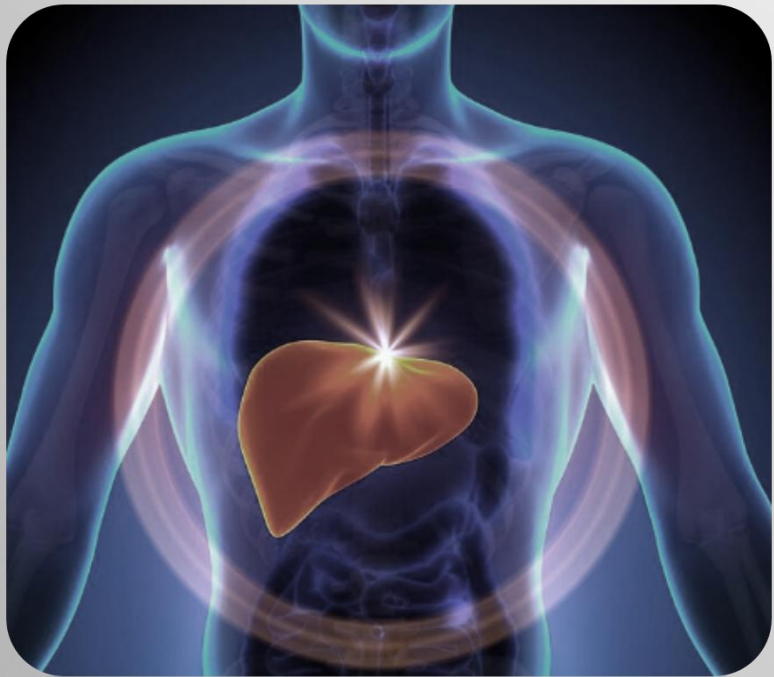
"Anyone who can stop or delay liver fibrosis would be able to cure most liver diseases"

----- Dr.Hans Popper
(USA)

Founder of modern Hepatology



Early stage of Liver diseases are hard to detect by patient

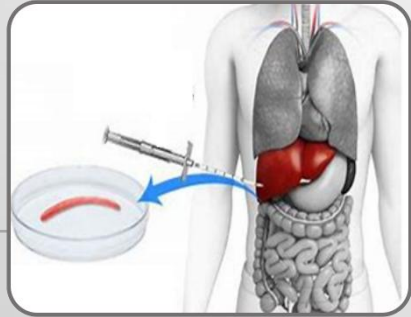


Chronic liver diseases — "Silence Killer"

- The liver is the organ in the body that has only few pain nerves*
- Chronic liver diseases have no obvious symptoms until develop into serious issues, such as hepatomegaly or pain in liver*



Existing clinical diagnosis methods ---Fibrosis



Biopsy

- **Advantage:** golden standard
- **Disadvantages:** *invasive* examination, poor repeatability, easy to cause complications, sampling error



Blood testing

- **Advantage:** repeatable
- **Disadvantages:** many markers with *low single specificity* and sensitivity

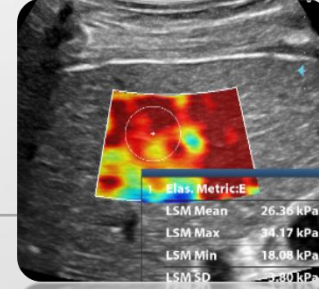


Existing clinical diagnosis methods ---Fibrosis



Imaging (US/CT/MRI)

- Advantage: non-invasive, easy follow up
- Disadvantages: **Unable to quantify and grade**, not sensitive to mild to moderate hepatic fibrosis



Imaging (Ultrasonic shear wave elastography)

- Advantage: Noninvasive, quantitative
- Disadvantages: **difficult to unify standards** due to different technologies of different manufacturers; high requirements for operation



Existing clinical diagnosis methods ---fatty liver



Ultrasound

- Only detects fat content >30%
- Advantage: cheap
- Disadvantage: **Low sensitivity**, not sensitive to early fatty liver; only qualitative judgment



MRI / CT

- MRI/CT: Fatty liver with more than 5% / 15% fat content can be detected
- Advantage: High sensitivity, MRI can be used for quantitative detection ;
- Disadvantage: **Expensive, Time cost**

*Non-invasive and
quantitative Liver
Evaluation*



The first choice for chronic liver disease examination

A

Noninvasive and
painless

To avoid liver biopsy, reduce the risk of biopsy, simple and fast, suitable for all kinds of people, non-invasive, safe and economic.

B

Sensitive and accurate

high sensitivity and accuracy to liver fibrosis and fatty liver in the early stage, which is conducive to the early screening, diagnosis, treatment and rehabilitation of liver diseases

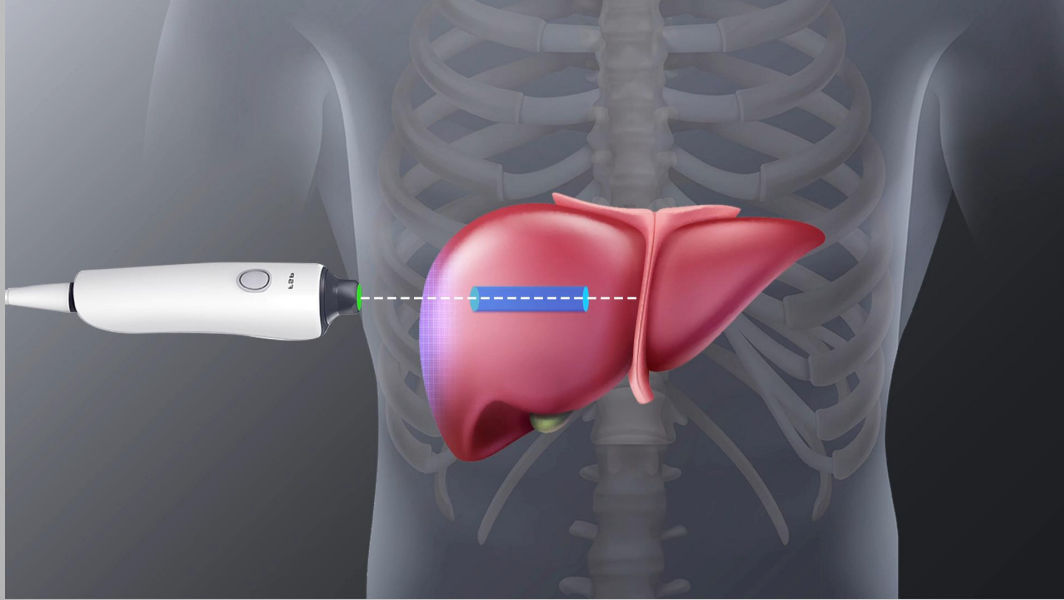
C

Quantitative and
reliability

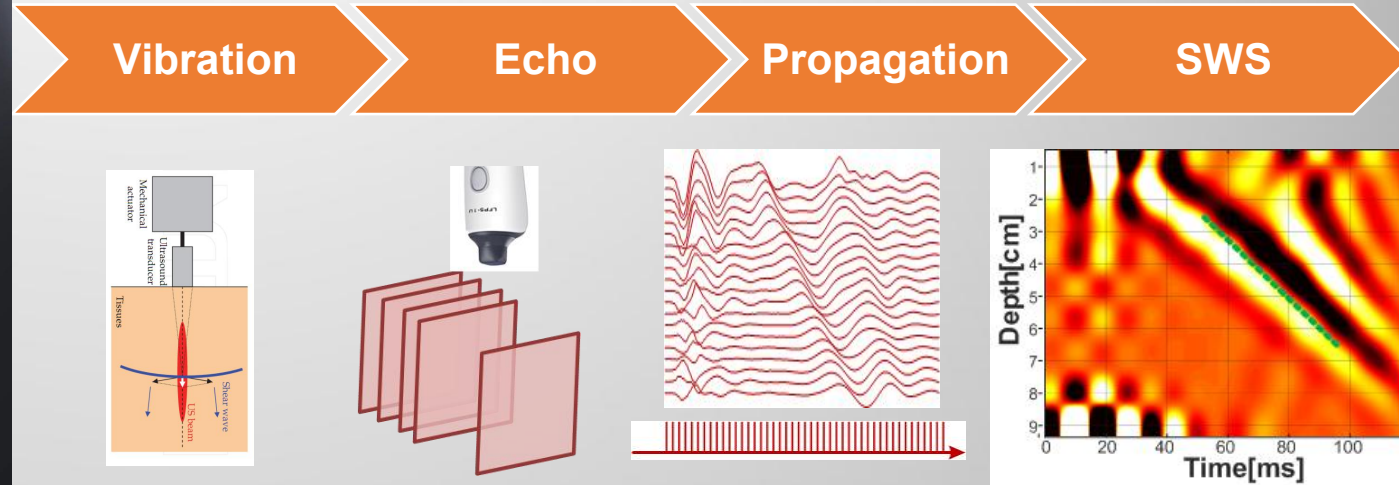
Quantitative detection and evaluation of liver fibrosis and steatosis degree, the technology has been widely recommended by the global liver disease guidelines, the diagnostic standard is mature and reliable.



Fibrosis evaluation



Vibrator generate the shear wave on the surface of body



- The harder tissue has smaller deformation under pressure $E(\text{kPa})$

$E(\text{kPa})$
Median
7.1

Fibrosis evaluation

- To evaluate the degree of hepatic fibrosis with reference to pathological staging criteria
- It can be classified quantitatively and is sensitive to mild to moderate hepatic fibrosis

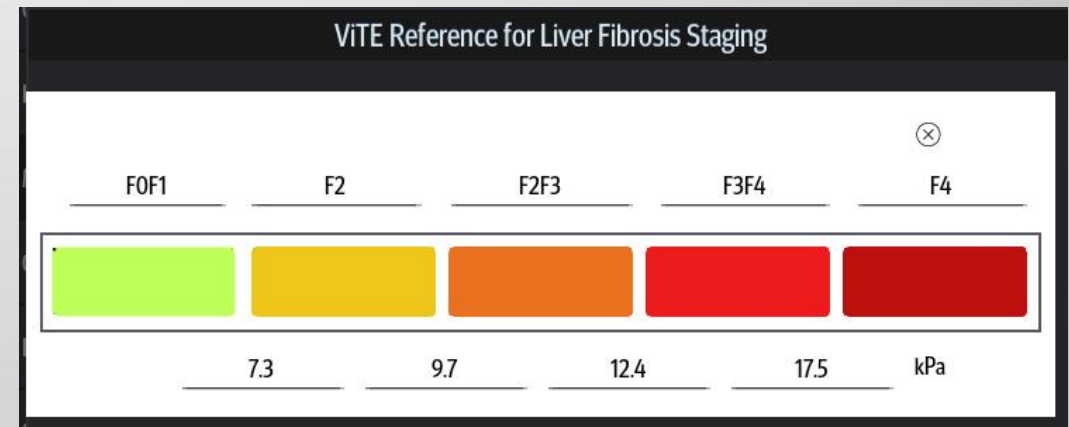
F0 No fibrosis

F1 Fibrosis but no fibrous septum formation

F2 A small amount of fibrous septum formation

F3 Massive fibrous septum without cirrhosis

F4 Cirrhosis



The examination results are interpreted by professional doctors in combination with patients' condition and other examination results

Fatty liver assessment

The basic principle of liver steatosis detection

When ultrasonic wave propagates in human body, it will be attenuated due to scattering and reflection

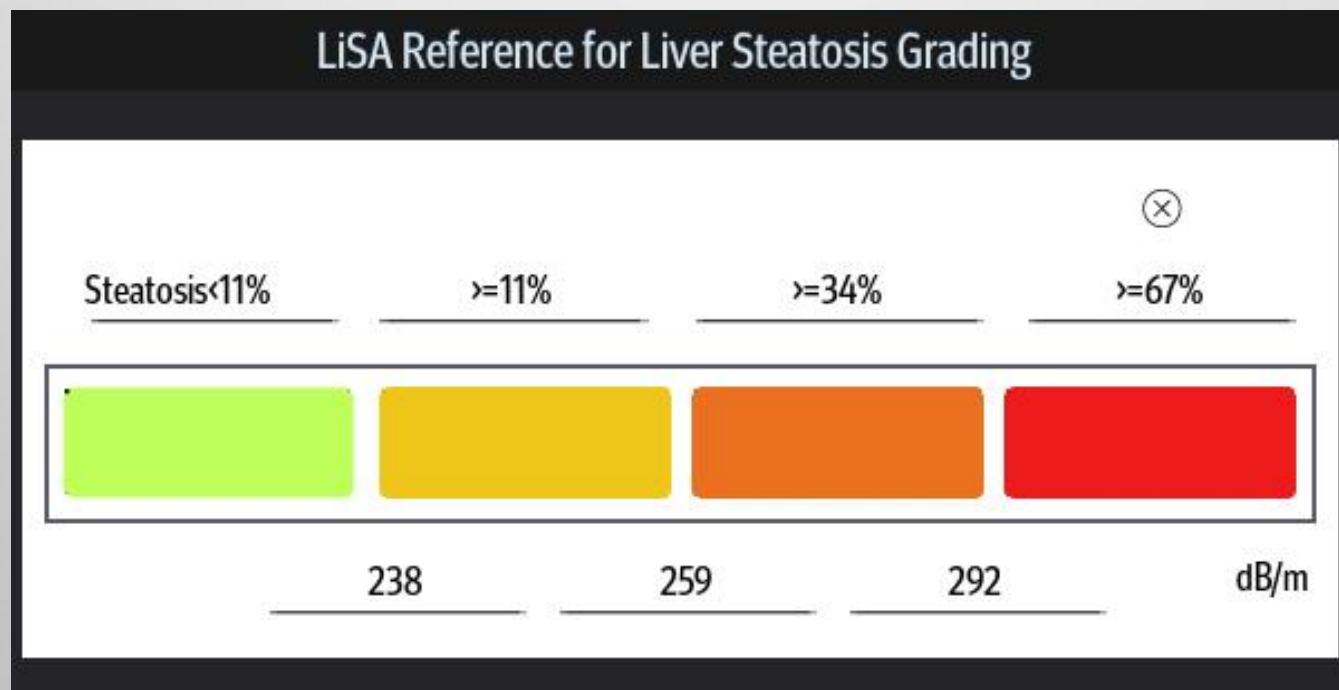
The attenuation of different tissues is different: the higher degree for fatty content of liver, the more severe for the attenuation of ultrasound





Fatty liver assessment

- To evaluate the degree of fatty liver with reference to pathological stage standard
- quantitative and sensitive to mild fatty liver



The examination results are interpreted by professional doctors in combination with patients' condition and other examination results

Quantitative Evaluation

E: Young's modulus



Quantitative analysis of liver stiffness

Provide quantitative detection of liver stiffness and liver fibrosis stage by transient elastography technology.

LiSA: **L**iver Ultra-**S**ound **A**ttenuation



Quantitative analysis of liver steatosis

The severity of liver steatosis can be quantitatively evaluated by LiSA technology.



Recommended by global guidelines for liver disease



AASLD



EASL



APASL



CMA



WFUMB

2010

- Recommendation of CMA guidelines for the prevention and treatment of chronic hepatitis B in China

2012

APASL guidelines for diagnosis and treatment of chronic hepatitis B
AASLD guidelines for the diagnosis and treatment of nonalcoholic fatty liver disease

2014

- AASLD chronic liver disease management recommendations

2016

- Recommended by APASL guidelines

2018

Recommended by guidelines for the prevention and treatment of nonalcoholic and alcoholic fatty liver disease in China
Consensus recommendation on diagnosis and treatment of nonalcoholic fatty liver disease and metabolic disorders in China

2009

- Recommended by AASLD guidelines for hepatitis C diagnosis and treatment in the United States

2011

- Recommended by clinical practice guidelines for chronic hepatitis C / B in EASL in Europe

2013

Recommended by expert's suggestion on standardization of diagnosis and treatment of fatty liver disease in China
China's expert opinion on the diagnosis of liver fibrosis by transient elastography

2015

WHO global guidelines for diagnosis and treatment of hepatitis C and B
Recommended by the European and Latin American Society of Hepatology clinical guidelines for noninvasive assessment of liver disease severity and prognosis
Recommendations for clinical use of ultrasound elastography of WFUMB (liver)
Recommended in the guideline for the prevention

2017

Recommended by practice guidelines for the diagnosis and management of nonalcoholic fatty liver disease in AASLD
Recommendations of Asia Pacific guidelines for the diagnosis and treatment of nonalcoholic fatty

2019

Consensus on diagnosis and treatment of liver fibrosis in China
Expert advice on standardization of diagnosis and treatment of fatty liver disease



Challenge of current products

Hard to position

Single element probe with vibrator, only M mode. Inexperienced operator or encounter difficult patient, the liver will be difficult to locate, and the success rate is greatly reduced

No quality control

No intuitive method to ensure the detection result is reliable.



Hard to operate

Large number of patient
↓
Each patient need to detect 10 group of data
↓
Each data is collected manually, lack of automatic solution

Expensive

Different types of probes need to be configured according to different body shapes. The cost of probe, calibration and maintenance will be very expensive.



Hepatus 5

*Non-invasive and quantitative
liver diagnose*

*New Vision for Liver
Care*



Hepatus solutions

Accurate

ViTE: Visualized
Transient Elastography



Convenient

Integrated probe design



Efficient

Q-Scan automatic
progress



Reliable

Pressure & Motion Stability
quality control index

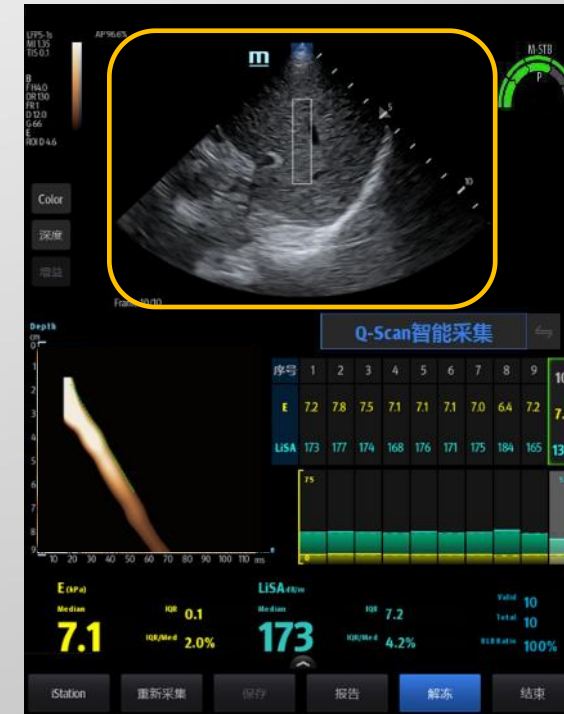




Accurate

New Tech: Advanced ViTE (Visualized Transient Elastography)

- Conduct quantitative detection and evaluation of liver fibrosis and steatosis under the visual guidance of real-time 2D ultrasound.
- Avoid blood vessels and lesions, improve the accuracy and reliability of the measurement
- Greatly reduce the operation difficulty, improve the accuracy and repeatability of quantitative measurements



Real time visibility
the whole process



Convenient

Integrated probe design

- Ultra-wideband frequency technology, it covers all patients of different ages and body shapes
- One probe for ultrasound visual guidance and quantitative detection, no need to switch probes during the scan
- The acquisition process can be accomplished with one integrated button



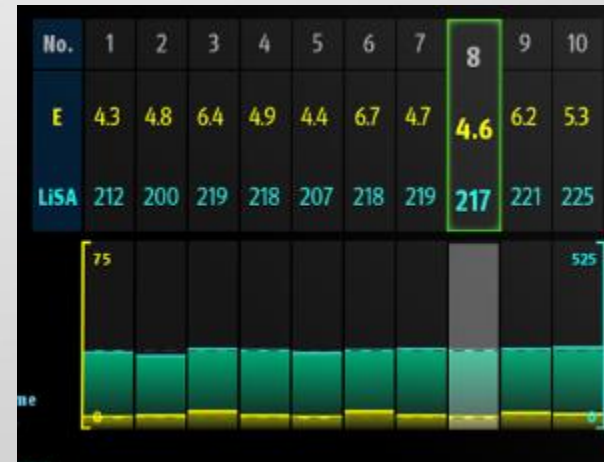


Efficient

Q-Scan intelligent acquisition

- 10 groups of effective data can be automatically and continuously acquired and intelligently screened by pressing one button
- Precise and valid detection results can be obtained immediately within merely **7 seconds** after positioning
- Greatly shorten the detection time and improve the efficiency

*Q-Scan unique
owned by Mindray*



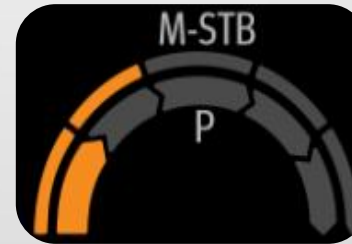


Reliable

Quality control index

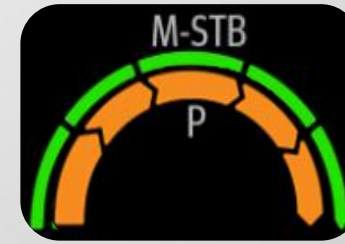
- Probe pressure index (P): quality control of probe pressure
- Motion Stability index (m-STB): ensure respiratory stability
- Clear indication with different color to improve the quality control during examination

*M-STB unique
owned by Mindray*



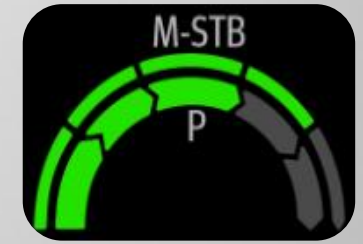
Respiratory
movement
disturbance

Probe pressure too
low



Stable respiratory
movement

Probe pressure too
high



Stable respiratory
movement

Proper probe
pressure





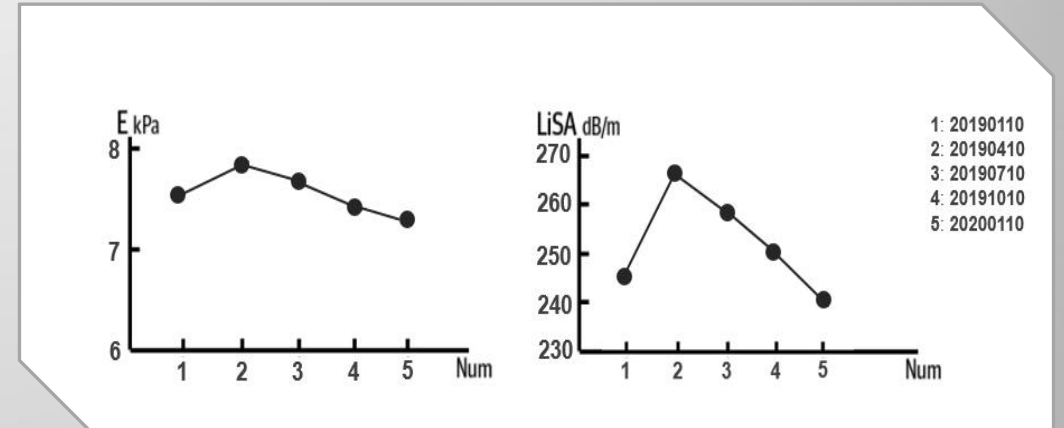
That's it?



Trend analysis, disease tracking

Intelligent trend analysis

- Follow up the exam results at later times and conduct trend analysis
- Convenient for doctors to monitor the changes of liver status, and conducive to disease tracking and treatment evaluation
- Detection data can be exported in batch for research and analysis





Outstanding experience

- Full touch screen
- Patients information input by Bar code scanning

- Built in battery for mobile / outdoor applications





New Vision for Liver Care





*Clinical applications and
approval*



Clinical applications and approval

Clinical examination and verification in hospitals

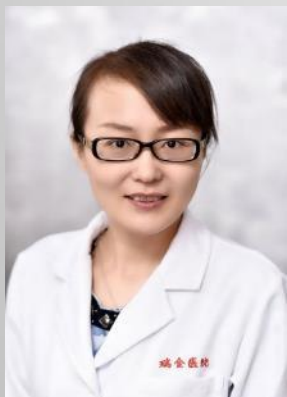
- Beijing Ditan Hospital Affiliated to Capital Medical University
- Ruijin Hospital Affiliated to Shanghai Jiaotong University School of Medicine
- Shenzhen Third People's Hospital Affiliated to South University of science and technology (National Center for medical research on infectious diseases)
- Eugenides Hospital, Greece





Clinical applications and approval

ViTE Expert evaluation



"ViTE With a visual real-time image guidance system, the operation is more convenient and the inspection success rate is higher. "

Professor Ren Xinping

Deputy chief physician of Ultrasound Medicine Department

Doctor of medicine, Fudan University

Ruijin Hospital Affiliated to Shanghai Jiaotong University

School of Medicine

"The (ViTE) probe works well and the B-mode image is sufficient for the guidance of the procedure"



Prof. Pavlos Zoumpoulis

Director of Interventional ultrasound department in Eugenides Hospital, Greece



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